

MUHAMMAD TAHA

AI/ML Engineer | Computer Vision | Multimodal AI | ML Systems | PyTorch · OpenCV · LLMs · Edge AI | NUST'26

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PROFESSIONAL SUMMARY

Computer Engineering graduate (NUST) specializing in Computer Vision, Multimodal AI, and ML Systems. Built and deployed real-time face anti-spoofing, document verification, and signature forgery detection pipelines during internship at TruID Technologies. Delivered SERENITY (Final Year Project), a local-first multimodal assistant integrating FER, SER, Whisper Tiny, RAG, quantized LLM inference, and persistent conversation memory for real-time therapeutic interaction. Built MedTraceAI for streaming clinical risk prediction with Kafka, FastAPI, Kubernetes, and Terraform. Combines strong AI modeling depth with production engineering, edge deployment, and full-stack execution.

TECHNICAL SKILLS

Computer Vision: OpenCV, MediaPipe, Face Anti-Spoofing, Document Verification, Signature Forgery Detection, Pose Estimation, 3D Reconstruction, Hough Transform, SIFT/ORB/SURF.

Deep Learning & Multimodal AI: PyTorch, TensorFlow Lite, CNN, BiLSTM, Attention, LSTM, Temporal Fusion Transformer, Whisper Tiny, Qwen2.5, RAG, FAISS, SHAP, XGBoost, LightGBM.

ML Systems & Backend: FastAPI, Kafka, Kubernetes, Terraform, SQLite, SQLAlchemy, Flask, REST APIs, Git, Linux.

Programming & Full-Stack: Python, TypeScript, C++, Java, SQL, Next.js 14, React, MySQL, JWT, Tailwind CSS.

Hardware & Embedded: Raspberry Pi 5, FPGA (Xilinx), Verilog HDL, ESP32, Arduino, Digital Logic Design, Custom ISA/Processor Design.

PROFESSIONAL EXPERIENCE

Computer Vision Intern | TruID Technologies

National Science and Technology Park (NSTP), Islamabad | Jul 2025 – Aug 2025

- Engineered face anti-spoofing pipelines for liveness detection across live-user, screen replay, and paper attack scenarios.
- Developed texture-based document verification models to classify original versus photocopied identity cards.
- Designed signature forgery detection models and optimized inference latency for real-time deployment.
- Integrated CV pipelines into production-oriented identity-security workflows.

Freelance Full-Stack Engineer | Self-Employed

Remote | Jan 2026 – Present

- Built a custom headless CMS for a dental practice and migrated a legacy platform to Next.js with real-time MySQL synchronization.
- Developed admin dashboard with drag-and-drop workflows, optimistic updates, and hybrid Vercel Blob + MySQL storage.
- Reduced content update turnaround by 90% (30+ minutes to under 2 minutes).
- Achieved sub-100kB initial load using SSR, caching, and AVIF/WebP optimization.

PROJECTS

SERENITY (Final Year Project): Smart Emotion Recognition and Neural Intervention

- Architected multimodal interaction loop combining facial emotion recognition (TFLite), speech emotion recognition (TFLite), Whisper Tiny transcription, and quantized Qwen2.5 response generation.
- Implemented RAG retrieval with FAISS and persistent context memory using SQLite + SQLAlchemy for personalized, context-aware responses.
- Built FastAPI backend with async orchestration, timeout guards, and graceful fallbacks to stabilize real-time inference workflows.
- Optimized for constrained hardware and designed architecture for hybrid edge-cloud offload evolution.

MedTraceAI: Real-Time Clinical Deterioration Prediction

- Developed streaming clinical prediction platform ingesting vitals, labs, medications, and ADT events for low-latency bedside risk scoring.
- Implemented LSTM and Temporal Fusion Transformer with multimodal feature fusion, calibration monitoring, and SHAP explainability.
- Deployed production architecture with Kafka, FastAPI, Kubernetes, and Terraform plus clinician-facing operational dashboards.

Full-Stack Dental Practice Platform and Custom Headless CMS

- Architected Next.js 14 + TypeScript + MySQL platform with custom admin dashboard and real-time content operations.
- Implemented hybrid storage with Vercel Blob and Git-backed assets to support scalable media + content workflows.
- Reduced update cycle by 90% and delivered sub-100kB initial load through SSR and caching strategies.

Human Pose Estimation and Classification

- Built and compared two pipelines on MPII: MediaPipe keypoint extraction versus from-scratch classical CV using contours, skeletonization, and geometric heuristics.
- Engineered joint-angle features and trained SVM/Random Forest classifiers with quantitative evaluation (PCKh@0.5, skeleton completeness, symmetry).

3D Environment Reconstruction from Multi-View Images

- Developed full multi-view reconstruction pipeline using SIFT/ORB/SURF correspondences, epipolar geometry, camera pose estimation, and stereo triangulation.
- Generated and refined sparse/dense point clouds for robotics, AR/VR, and spatial scene understanding use cases.

Real-Time Image Analysis for Self-Driving Capabilities

- Built edge vision system on Raspberry Pi 5 for lane detection, obstacle segmentation, and directional decision output.
- Optimized pipeline using ROI selection, downsampling, adaptive thresholding, edge detection, and Hough transforms for real-time performance.

Lung and Colon Cancer Classification

- Developed CNN-based histopathology classifier for 5 cancer classes and achieved 98.6% validation accuracy through architectural tuning.

Santander Customer Transaction Prediction

- Built tabular ML pipeline with feature engineering and model benchmarking across Logistic Regression, Random Forest, XGBoost, and LightGBM.
- Applied SHAP and permutation importance for interpretable prediction analysis on high-dimensional financial data.

FPGA Snake Game with Hardware-Implemented Ghost AI

- Engineered processor-less game engine on Xilinx FPGA using Verilog HDL with VGA output at 640x480 @ 60Hz.
- Implemented autonomous ghost-tracking AI directly in hardware using Manhattan distance minimization logic.

Custom 16-bit Harvard-Architecture Processor

- Designed a complete 16-bit CPU in Verilog with custom ISA, control unit, ALU, datapath, and register file.

LEADERSHIP & EXTRACURRICULAR

Lead of Human Resources (HR) | COMPPEC (Computer Project Exhibition Competition) (Apr 2024 – May 2026)

- Managed recruitment and coordination of volunteers to ensure smooth execution of the university's largest project exhibition.

Class Representative | NUST (Sep 2024 – May 2026)

- Served as primary liaison between faculty and students, resolving academic concerns and coordinating class schedules.

Event Management | BurRaq (NUST Debating Society) (Nov 2022 – Sep 2024)

- Organized debating events and declamation contests, managing logistics and participant engagement.

Awards

- Winner: Declamation Competition, BurRaq Extempore Competition (Nov 2022).

EDUCATION

Bachelor of Computer Engineering | National University of Sciences and Technology (NUST) | Islamabad, Pakistan

2022 – 2026 | Relevant Coursework: Data Structures, Operating Systems, Computer Networks, Computer Architecture, Digital Image Processing, DSP, AI, Machine Learning, Computer Vision, Software Engineering, Database Engineering.

CERTIFICATIONS

- Deep Learning Specialization — DeepLearning.AI (Dec 2025)
- Machine Learning Specialization — Stanford Online (Jul 2025)
- AI for Everyone — DeepLearning.AI (Jul 2024)
- Introduction to Front-End Development — Meta (Oct 2023)
- CS50: Introduction to Programming with Python — Harvard University (Sep 2023)